

Multiple Choice

1. **Answer:** C
Options: A) linear in D; B) linear in N; C) polynomial in D; D) dependent on the number of iterations
Note: Correct option: C - polynomial in D
2. **Answer:** D
Options: A) It can not be applied to non-differentiable functions.; B) It can not be applied to non-continuous functions.; C) It is hard to implement.; D) It runs reasonably slow for multiple linear regression.
Note: Correct option: D - It runs reasonably slow for multiple linear regression.
3. **Answer:** A
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: A - True, True
4. **Answer:** D
Options: A) 3; B) 4; C) 7; D) 15
Note: Correct option: D - 15
5. **Answer:** D
Options: A) Partitioning based clustering; B) K-means clustering; C) Grid based clustering; D) All of the above
Note: Correct option: D - All of the above

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '2'.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: ' $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$ '.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, True'.

Long Form Response

11. **Answer:** `def count_Char(str,x): | return count`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def dif_Square(n): | return True | return False`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key